**NUST Institute of Civil Engineering (NICE-SCEE)**
*National University of Sciences & Technology (NUST)*

# CS114: Fundamentals of Programming

## Class: BECE2018

## Lab 06: Relational, Logical and Conditional Operators in C++

**Instructor: Qurrat-ul-ain Babar**

# Lab 06: Relational, Logical and Conditional Operators in C++

The aim of this lab is to cover some other basic operators of C++ programming.

Our aims today are:

To learn

- Relational Operators
- Logical Operators
- Conditional Operators

## Relational Operators

- In order to evaluate a comparison between two expressions
- The result of a relational operation is a Boolean value i.e. True or False.
- Here is a list of the relational and equality operators that can be used in C++

There are following relational operators supported by C++ language

Assuming variable **A** holds **10** and variable **B** holds **20**, then

| Operator | Description | Example |
|---|---|---|
| == | Checks if the values of two operands are equal or not, if yes then condition becomes true. | (A == B) is not true. |
| != | Checks if the values of two operands are equal or not, if values are not equal then condition becomes true. | (A != B) is true. |
| > | Checks if the value of left operand is greater than the value of right operand, if yes then condition becomes true. | (A > B) is not true. |
| < | Checks if the value of left operand is less than the value of right operand, if yes then condition becomes true. | (A < B) is true. |

| | | |
|---|---|---|
| >= | Checks if the value of left operand is greater than or equal to the value of right operand, if yes then condition becomes true. | (A >= B) is not true. |
| <= | Checks if the value of left operand is less than or equal to the value of right operand, if yes then condition becomes true. | (A <= B) is true. |

## Logical Operators

While relational (comparison) operators can be used to test whether a particular condition is true or false, they can only test one condition at a time. Often we need to know whether multiple conditions are true at once. For example, to check whether we've won the lottery, we have to compare whether all of the multiple numbers we picked match the winning numbers. In a lottery with 6 numbers, this would involve 6 comparisons, *all* of which have to be true. Other times, we need to know whether any one of multiple conditions is true. For example, we may decide to skip work today if we're sick, or if we're too tired, or if won the lottery in our previous example. This would involve checking whether *any* of 3 comparisons is true.

Logical operators provide us with this capability to test multiple conditions.

| Operator | Description | Example |
|---|---|---|
| && | Called Logical AND operator. If both the operands are non-zero, then condition becomes true. | (A && B) is false. |
| \|\| | Called Logical OR Operator. If any of the two operands is non-zero, then condition becomes true. | (A \|\| B) is true. |
| ! | Called Logical NOT Operator. Use to reverses the logical state of its operand. If a condition is true, then Logical NOT operator will make false. | !(A && B) is true. |

# Conditional Statement (if & if-else Statement):

- The "if statement" is used to execute a set of statements after testing a condition.
- "if-else statement" have two part i.e. TRUE and FALSE
- If the condition is true then the TRUE part executes and if the condition is false then the FALSE part executes.

*Syntax:*
*If you have a single statement that needs to be executed if the condition is true then the syntax is as follows:*

```
if(condition)
      statement;
```

*If you have multiple statements that needs to be executed if the condition is true then the { } are used to group the statements and the syntax becomes as follows:*

```
if(condition)
{
      statement1;
      statement2;
      …
      …
      statement(n);
}
```

```
if(condition)
{
      statement1;
      …
      …
      statement(n);
}
else
{
      statement1;
      …
      …
      statement(n);
}
```

# Deliverables

1. Execute all **Tasks** given in this document and take screen shots of the code and outputs of each task.

Upload single word document on LMS. Follow the naming convention for file. No late submissions will be accepted.

Deadline: Check deadline of the submission link on LMS.

## Task 1

Write a program to input a number and show that the input number is EVEN or ODD (using if statement and conditional operation)

## Task 2

Write a program to input two numbers and show the greater number (using conditional operator).

## Task 3

Write a piece of code that takes three integer num1, num2 and num3 having values 20, 30 and 30 respectively. At one time compare two integers and check whether the two variables compared are equal or not.

## Task 4

Write a program to input three integer values. Compare the numbers and show the highest number or lowest number

## Task 5

Write a program by taking two float values and show the average of two numbers.

## Task 7

Write a program that takes 5 numbers as input from user and prints the odd numbers on output screen.

# Task 8

Write a program that takes first 10 natural numbers and prints the sum of even numbers.